UNIVERSIDAD TÉCNICA NACIONAL

DEPARTAMENTO DE MATEMÁTICA

ME-003 – CÁLCULO I

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PORTAFOLIO DE EVIDENCIAS – ENTREGA #2

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CARRERA: Ingeniería del Software

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Portafolio #2

1) $\int \frac{(3-4x)^2 (x-5)}{\sqrt{x}} dx$

Fórmula notable

$(a+b)^2 = a^2 + 2ab + b^2$

$(3-4x)^2 = 3^2 - 2.3.4x + 4x^2$

$(3-4x)^2 = 9 - 24x + 16x^2$

Multiplico

$(x-5)(9-24x+16x^2)$

$(9x - 24x^2 + 16x^3 - 45 + 120x - 80x^2)$

$16x^3 - 104x^2 + 129x - 45$

$= \int \frac{16x^3 - 104x^2 + 129x - 45}{\sqrt{x}} dx$

$= \int \left(\frac{16x^3}{\sqrt{x}} - \frac{104x^2}{\sqrt{x}} + \frac{129x}{\sqrt{x}} - \frac{45}{\sqrt{x}} \right) dx$

$= \int \left(16x^{5/2} - 104x^{3/2} + 129x^{1/2} - 45x^{-1/2} \right) dx$

$= 16 . \frac{x^{7/2}}{7/2} - 104 . \frac{x^{5/2}}{5/2} + 129 \frac{x^{3/2}}{3/2} - 45 . \frac{x^{1/2}}{1/2} + C$

$= 16 . \frac{2}{7} x^{7/2} - 104 . \frac{2}{5} x^{5/2} + 129 . \frac{2}{3} x^{3/2} - 45 . \frac{2}{1} x^{1/2} + C$

$$\boxed{= \frac{32}{7} x^{7/2} - \frac{208}{5} x^{5/2} + 86 x^{3/2} - 90 x^{1/2} + C}$$

2) $\int \sqrt[3]{x}\left(2x+\frac{1}{x}-5x^2\right)dx$

$= \int x^{1/3}\left(2x+\frac{1}{x}-5x^2\right)dx$

$= \int \left(2x^{4/3}+x^{-2/3}-5x^{7/3}\right)dx$

$= 2.\frac{x^{7/3}}{7/3}+\frac{x^{1/3}}{1/3}-5.\frac{x^{10/3}}{10/3}+C$

$= 2.\frac{3x^{7/3}}{7}+3.x^{1/3}-5.\frac{3x^{10/3}}{10}+C$

$$\boxed{= \frac{6}{7}x^{7/3}+3x^{1/3}-\frac{3}{2}x^{10/3}+C}$$